

The Blue Economy and Its Efficiency in Enhancing Food Security in Algeria during the Period 2010–2022: The Case of the Fisheries and Aquaculture Sector

الاقتصاد الأزرق وكفاءته في تعزيز الأمن الغذائي بالجزائر خلال الفترة 2010–2022: قطاع الصيد البحري والتربية المائية نموذجا

Denden Fethi Hassen

University of Mostaganem (Algeria), fethihassen.denden@univ-mosta.dz

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Abstract:

This study aims to assess the role of the blue economy in enhancing food security by addressing the concepts related to the blue economy and food security, and by analyzing the efficiency of the fisheries sector as a model of the blue economy in strengthening food security in Algeria.

The study concludes that the sector has been inefficient in promoting food security, as it has not achieved productive and allocative efficiency. It failed to bridge a fish consumption gap of 17 kilograms per capita annually and did not utilize more than 40% of the fishing fleet. Furthermore, it has been inefficient in providing affordable fish prices, which increased by 200%, and in creating additional employment opportunities, with its contribution to total employment not exceeding 1.21% of the overall labor force in the economy.

Keywords: Blue Economy, Food Security, Fisheries, Fish Production, Aquaculture.

JEL Classification Codes: Q22, Q18, O13, Q57.

¹ *Corresponding author: Denden Fethi Hassen, e-mail: fethihassen.denden@univ-mosta.dz.*

ملخص:

تهدف هذه الدراسة إلى تقييم دور الاقتصاد الأزرق في تعزيز الأمن الغذائي، حيث تم التطرق إلى المفاهيم المرتبطة بالاقتصاد الأزرق والأمن الغذائي وتحليل كفاءة قطاع الصيد البحري كنموذج للاقتصاد الأزرق في تعزيز الأمن الغذائي بالجزائر.

خلصت الدراسة إلى عدم كفاءة القطاع في تعزيز الأمن الغذائي حيث لم يتميز بالكفاءة الإنتاجية والتخصيصية لعدم قدرته على تغطية فجوة غذائية من السمك قدرها 17 كلف للفرد سنوياً وعدم استغلاله لأكثر من 40% من الأسطول السفن الصيد. وعدم كفاءته في توفير أسعار مناسبة للأسماك، إذ ارتفعت بنسبة 200%. وعدم كفاءته في توفير مزيد من فرص العمل حيث لم تتجاوز نسبة مساهمته في التشغيل 1.21% من إجمالي العمالة في الاقتصاد.

كلمات مفتاحية: الاقتصاد الأزرق، الأمن الغذائي، الصيد البحري، الانتاج السمكي، التربية المائية.

تصنيفات JEL: Q57، O13، Q18، Q22

1. INTRODUCTION

In light of the rapidly accelerating pace of global demographic growth, it has become imperative for the economies of the world to strike a balance between individual well-being and the rational use of scarce natural resources. This has led to the emergence of the blue economy, which seeks to ensure the sustainability of water resources and the economies that depend on them. The blue economy is valued at approximately USD 1.5 trillion annually and provides a livelihood for more than 820 million people worldwide. Among the most important sectors of the blue economy, which requires sustainable management of scarce water resources, is the fisheries and aquaculture sector.

At the beginning of the 21st century, the world witnessed increasing growth in the fisheries and aquaculture sector. Global aquatic animal production rose to 223.2 million tons in 2022, the highest level ever recorded, with a market value of USD 472 billion. The sector created around 62 million jobs in the primary production chain of aquatic animals and provided an average of 20.7 kilograms of fish per capita per year

globally, accounting for about 15% of animal protein intake. This underscores its significant role in contributing to the achievement of global food security.

Given the economic, social, and security importance of food security, it has become necessary for countries around the world, including Algeria, to devote great attention to this issue by adopting economic policies and development programs for the fisheries and aquaculture sector. Algeria possesses significant potential and natural resources in this sector that enable it to make an effective contribution to enhancing food security, as it has a coastline of 1,641.32 km stretching from the western to the eastern borders.

In light of these considerations, the following research question arises:

To what extent has the fisheries and aquaculture sector, as a model of the blue economy, contributed to enhancing food security in Algeria during the period 2010–2022?

1.2 Research Hypothesis: In light of the research problem raised, we propose the following hypothesis: given the considerable potential in terms of coastline length and the number of fishing ports, the fisheries and aquaculture sector is characterized by sufficient efficiency in enhancing food security in Algeria.

1.3 Research Objectives: This study seeks to achieve the following objectives:

- To provide an overview of the blue economy, its activities, characteristics, and its role in achieving security.
- To highlight the role played by the fisheries and aquaculture sector in enhancing food security in Algeria.
- To shed light on the extent to which the fisheries and aquaculture sector is efficient in contributing to food security in Algeria.

1.4 Significance of the Study: The significance of this study lies in the role of food security in achieving the economic and social well-being of individuals, which Algeria places utmost importance on ensuring, in addition to demonstrating the contribution of the blue economy in achieving food security.

1.5 Research Methodology: Given the nature of the study and in order to achieve its intended objectives, the descriptive and analytical approach was adopted. This involved addressing the various concepts and theoretical framework related to the blue economy and food security, while highlighting the role of the fisheries sector in Algeria as one of the key components of the blue economy in enhancing food security, based on tables, data, and analytical evidence.

2. Theoretical Framework of the Blue Economy

2.1 Definition of the Blue Economy: The Belgian economist Gunter Pauli introduced the concept of the blue economy, defining it as: an economy that emphasizes the preservation and sustainable management of water resources in order to sustain the economies that rely on them, based on the assumption that healthy ocean ecosystems are more productive and are essential for the sustainability of ocean-based economies. (Ben Abbes & Laouar, 2019, p. 364)

The World Bank says that the blue economy is made up of all the sectors, economic activities, and policies that work together to make sure that water and marine resources are used in a way that is good for the environment. The goal of this idea is to promote economic growth, enhance and protect people's livelihoods, and make sure that marine and aquatic habitats are used in a way that is sustainable.. (World Bank, 2017).

The Food and Agriculture Organization (FAO) of the United Nations defines the blue economy as an economy that is based on investing in marine activities across various water bodies—oceans, seas, rivers, lakes, and others—in order to achieve optimal returns from them, such as fisheries, maritime transport, logistics, and tourism services. The FAO launched what it termed the Blue Growth Initiative, which aims to protect different water bodies from growing threats such as illegal fishing and pollution. It further emphasized adherence to the principles of sustainable development through the optimal and rational use of water resources, while promoting cooperation among countries in this regard. Moreover, FAO highlighted the importance of fostering innovation, development, and investment to support food security, enhance the sustainable management of

water resources, and reduce poverty. (Ben Abbes & Laouar, 2019, p. 365)

2.2 Objectives of the Blue Economy: The blue economy seeks to ensure the sustainability of water resources for future generations with the aim of achieving the following objectives: (Ben Abbes & Laouar, 2019, pp. 366-367)

- Rationalizing natural fish harvesting in order to restore and preserve fish stocks by preventing harmful and illegal fishing activities.
- Promoting and enhancing innovation and investment to contribute to food security and the sustainable management of marine resources.
- Reducing poverty and hunger by creating job opportunities, thereby improving individuals' living standards.
- Enhancing the sustainability of marine resources by planting marine vegetation, protecting aquatic life, and supporting sustainable fish production activities.
- Strengthening regulatory systems through the provision of ecosystem services, restoring vital coastal habitats, and preserving biodiversity.
- Preserving marine, fishery, natural, and water resources.

2.3 Components of the Blue Economy:

The blue economy includes many different industries and activities that depend on water resources. The following are the key sectors that represent the main components of the blue economy: (Meskine & Zergout, 2019, pp. 63-64)

2.3.1 Fisheries and Aquaculture: Fisheries are one of the most essential parts of the blue economy since they help the country make more money, make sure people have enough food, and create jobs.

2.3.2 Marine Biotechnology: Marine biotechnology contributes to production across numerous economic sectors. It employs advanced technologies in the production of cosmetics, pharmaceuticals, various chemicals, nutritional supplements, and enzymes, making it one of the fastest-growing activities within the blue economy.

2.3.3 Marine Renewable Energy: Wave energy, tidal energy, and ocean thermal energy are all kinds of marine renewable energy, offshore solar, and offshore wind, the latter being the most widespread worldwide. Marine renewable energy accounts for nearly 22% of total global energy.

2.3.4 Maritime Transport and Logistics: Maritime transport is a primary means of international trade. The UN Conference on trade and Development (UNCTAD) says that marine commerce made up more than 70% of all trade in 2016.

2.3.5 Marine Industries: Marine industry involves various economic sectors such as the construction, maintenance, and repair of boats, ships, and even submarines, as well as other floating structures. It also includes marine technologies, ship recycling, and dismantling. This activity represents an important component of the blue economy.

2.3.6 Trade, Communication, and Information Technology: This includes maritime trade, marine insurance services, and ship financing, in addition to marine information and communication technologies, environmental and marine engineering consultancy, meteorological services, project management, and submarine and ship communication networks.

2.3.7 Offshore and Deep-Sea Mining: This refers to the extraction of natural resources found in seas and oceans, such as gas, oil, gold, and diamonds, as well as sulfur, sand, and gravel. Many countries rely heavily on this activity.

2.3.8 Marine and Recreational Tourism: The vast maritime areas around the world provide numerous opportunities for tourism and recreational services, including surfing, diving, coastal tourism, and coral reef exploration. This sector plays a vital role in the blue economy.

2.4 Principles of the Blue Economy: The World Wide Fund for Nature (WWF) has outlined a set of key principles for the blue economy, the most important of which are: (Gharbi & Mohammedi, 2019, pp. 78-79)

- Restoring, conserving, and protecting biodiversity, functionality, productivity, and the value of marine ecosystems, along with the natural benefits upon which their prosperity depends.
- Relying on renewable energy, clean technologies, and material recycling to ensure continuous economic and social sustainability.
- Managing the blue economy through governance approaches that are inclusive, sustainable, participatory, proactive, and well-informed.

- Promoting dialogue with stakeholders while establishing a clear reference framework for the blue economy.
- Emphasizing commitment from governments and all stakeholders to a sustainable blue economy vision, and contributing to translating this vision into tangible reality.
- Engaging stakeholders in decision-making processes related to the blue economy through educational frameworks and awareness-raising initiatives.
- Guiding decision-making in both the public and private sectors with respect to the blue economy.

3. Theoretical Framework of Food Security

3.1 The Concept of Food Security: Food security is described as the ability of a society to supply its members with enough food in both the short and long term, in terms of quantity and quality, and at prices that are commensurate with their income levels. (Hamdan, 1999, p. 16). From this definition, it can be inferred that food security reflects the capacity of societies to ensure food provision—either through domestic production or imports—at appropriate quality, quantity, and prices for all members of society.

This current definition contrasts from the previous idea of food security, which was largely tied to establishing self-sufficiency by depending on a country's own resources to cover its food demands purely via local production, without turning to imports (Massoud, 2022, p. 9)

The concept of food security has since evolved to become one of the pillars of comprehensive national security. This does not necessarily imply that a country must possess vast natural resources, as having such resources alone is insufficient to achieve food security, nor is possessing financial resources by itself adequate. Instead, food security is tied to the consistent provision of sufficient, nutritious, and affordable food to all segments of society (Ahmed, 2014, p. 19).

The FAO adopted the definition agreed upon at the World Food Summit (1996), which states: Food security occurs when all people, at all times, have physical and economic access to sufficient, secure, and nutritious food that matches their dietary needs and food choices for an

active and healthy life. FAO noted that this definition encompasses multiple dimensions of food security (FAO, 2006, p. 1).

The Arab Organization for International Development further concluded that food security means producing more food than the country requires, in an economically efficient and balanced manner, where domestic food production enjoys a competitive advantage enabling it to compete with foreign production in international markets (Menad & Zerouat, 2025, p. 219).

Despite the diversity of definitions, food security remains closely tied to the values and culture of societies, varying from one society to another. Nevertheless, there is consensus on its overarching importance and its deep connection to these societal values (Khemissi & Benkheznadji, 2019, p. 141).

3.2 Dimensions of Food Security: In a report published by the FAO, four key dimensions of food security were identified (FAO, 2006, p. 1):

- **Availability:** Ensuring that sufficient amounts of food of suitable quality are supplied through domestic production, imports, or food aid.
- **Access:** The ability of individuals to acquire adequate resources to obtain appropriate foods for a healthy diet.
- **Utilization:** The proper use of food to achieve individual nutritional well-being and meet physiological needs, which requires an adequate diet, clean and safe drinking water, healthcare, and sanitation.
- **Stability:** The capacity to get adequate food at all times, without this access being transitory or disturbed by unexpected shocks (such as economic or climatic crises) or repeated events (such as seasonal food insecurity).

4. The Efficiency of the Fisheries and Aquaculture Sector as a Model of the Blue Economy in Achieving Food Security in Algeria

Algeria possesses significant potential in fisheries and aquaculture, with a coastal area stretching along a 1,641.32 km shoreline and an estimated fish stock of about 6 million tons. Since 2008, Algeria has launched a plan for fish egg hatching through two pilot projects in the

provinces of Sétif and Sidi Bel Abbès to produce fish fry, which strengthened the country's orientation toward aquaculture activities and the development of several species of freshwater fish as part of a strategy to promote the fisheries sector.

The successful breeding of 6,800 tilapia and carp eggs in 13 ponds in Algiers and its surroundings encouraged investors by demonstrating the feasibility of expanding the practice on a larger scale. This initial experiment in aquaculture contributed to its dissemination across other coastal provinces. (Ben Abbes & Laouar, 2019, pp. 374-75)

4.1 The Contribution of Marine Fisheries and Aquaculture Production to Enhancing Food Security in Algeria: We have previously noted that achieving food security requires ensuring the availability of food in adequate quantities, of appropriate quality, and at affordable prices for all members of society. Accordingly, attaining this goal depends either on increasing domestic production or on enhancing individuals' purchasing power, enabling them to access the necessary food. In this context, the marine fisheries and aquaculture sector in Algeria plays a central role in maintaining this balance, as it contributes to the supply of fish in sufficient quantities and at reasonable prices, in addition to generating new economic opportunities that improve individual income and strengthen purchasing power.

4.1.1 The Role of Marine Fisheries and Aquaculture in Supplying Fish Food: A fundamental role of the marine fisheries and aquaculture sector lies in ensuring the provision of sufficient quantities of fish protein to meet individuals' dietary needs. Therefore, this section examines the development of fish production in Algeria and the extent to which it satisfies the population's needs, using two key indicators: the annual per capita share of fish production and the annual per capita share of apparent fish consumption (in kilograms). The following table illustrates this.

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Table 1. Evolution of Fish Production, Annual Per Capita Fish Production, Annual Per Capita Apparent Consumption, and Fishing Fleet during the Period 2010–2022

Years	Fish Production (tons)	Annual Per Capita Fish Production (kg)	Annual Per Capita Apparent Consumption (kg)	Marine Fishing Fleet
2010	95 168	2,65	3,31	4191
2011	104 008	2,83	3,59	4327
2012	108 207	2,89	3,80	4414
2013	102 220	2,67	3,54	4580
2014	100 150	2,56	3,68	4787
2015	105 200	2,63	3,67	5034
2016	102 140	2,5	3,53	5333
2017	108 300	2,6	3,52	5494
2018	120 354	2,83	3,45	5617
2019	104 880	2,42	3,18	5784
2020	87 622	1,98	2,72	5853
2021	97 508	2,17	2,91	5 966
2022	116 790	2.56	3.15	6044

Source: Compiled by the researcher based on:

1. Office National des Statistiques, Les principaux indicateurs du secteur de la pêche 2010–2022, <https://www.ons.dz/spip.php?rubrique263>
2. Office National des Statistiques, Démographie Algérienne 2020–2023, https://www.ons.dz/IMG/pdf/Demographie_Alg2020_2023.pdf

From Table 1, it is observed that fish production fluctuated during the period (2010–2022) without achieving sustained growth, with an average production of 104.04 thousand tons. During this period, fish production increased by 13%, rising from 95,168 tons to 108,207 tons. This growth had a positive impact on the annual per capita share of production, which increased by 240 grams, reaching 2.89 kg compared to 2.65 kg previously. Similarly, the annual per capita share of apparent consumption rose by 490 grams.

However, production declined during the period 2012–2014 by 7.44%, reaching around 100 thousand tons, a decrease of approximately 8 thousand tons. This drop led to a reduction of 330 grams in the annual per capita share of production. In response, the government was compelled to increase fish imports from 30 thousand tons in 2013 to 45 thousand tons in 2014,

with the aim of improving the annual per capita share of apparent consumption by 140 grams during the period (2010–2013).

Fish production improved significantly during the period 2014–2018, reaching 120 thousand tons, recording a growth rate of 20.17%, equivalent to an increase of 20 thousand tons. This improvement was accompanied by a slight increase of 270 grams in the annual per capita share of production. However, the annual per capita share of apparent consumption declined to 3.45 kg, due to a reduction in imports to 30 thousand tons in 2018, compared to 45 thousand tons in previous years.

During the period 2018–2022, fish production experienced a notable decline, reaching its lowest level in 2020 with 87.62 thousand tons. Consequently, the annual per capita share of production dropped to 1.98 kg, a decrease of 850 grams compared to 2018. Production then witnessed a slight recovery, reaching 116.79 thousand tons in 2022, with the annual per capita share of fish production improving to 2.56 kg.

4.1.2 The Efficiency of the Marine Fisheries and Aquaculture Sector in Improving Production

Table 01 shows that, despite the continuous increase in the number of fishing vessels from 4,191 in 2010 to 6,044 in 2022 an increase of 42% production did not follow the same growth pattern, rising by only 2.4% over the same period. This discrepancy raises questions regarding the efficiency of the marine fisheries sector in improving production, as efficiency reflects the capacity to optimally utilize available resources and achieve the desired objectives.

4.1.2.1 Production Efficiency in the Optimal Utilization of Resources: According to the annual reports of the Office National des Statistiques on the main indicators of the fisheries sector, the proportion of inactive vessels in the fishing fleet ranged between 40% and 44.4% during the period (2018–2022). This confirms that more than 40% of the physical capital in the marine fisheries and aquaculture sector is not utilized within the production chain. Consequently, the sector suffers from low efficiency in the optimal use of available economic resources, which negatively impacts production, leading to stagnation in most periods. These findings highlight the need to adopt effective strategies to improve vessel utilization and

enhance productivity in the sector.

4.1.2.2 Production Efficiency in Ensuring Adequate Supply: The fisheries sector aims to enhance food security by providing sufficient quantities of fish to meet the needs of the population. The average annual per capita fish production during the period (2010–2022) was 2.56 kg, whereas the average annual per capita apparent fish consumption stood at 3.39 kg over the same period. This reflects a food gap between production and consumption, largely attributed to continuous imports, which totaled over 470 thousand tons during the study period, with an annual average of 36.15 thousand tons, representing approximately 34.75% of the sector's average annual production.

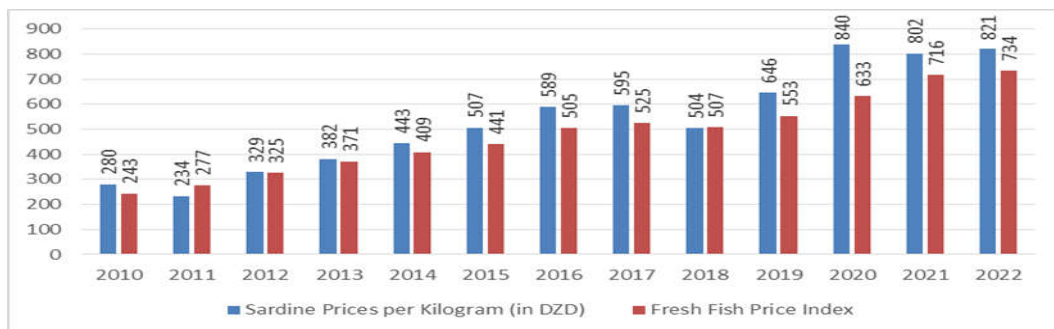
When compared to the global average annual per capita apparent fish consumption of 20.25 kg during the period (2010–2022) (Food and Agriculture Organization , 2024, p. 4), a significant food gap becomes evident. The shortfall in per capita fish consumption is estimated at about 16 kg annually, while the deficit in per capita fish production amounts to 17.54 kg annually. This means that the global per capita consumption exceeds local consumption by more than five times, while global per capita production surpasses its local counterpart by more than seven times. This situation underscores the need to strengthen domestic production in order to bridge this gap and reduce dependence on imports.

These figures indicate that the fisheries sector covers only 12.73% of individual fish consumption needs, reflecting weak production relative to domestic demand. This situation has compelled the government to rely on continuous imports to narrow the gap between production and consumption, totaling over 470 thousand tons during the study period, with an annual average of 36.15 thousand tons, equivalent to 34.75% of the sector's average annual production. This confirms the pressing need to develop strategies aimed at improving productivity and ensuring sustainability in the fisheries sector.

4.3 The Contribution of the Fisheries Sector to Providing Food at Affordable Prices: Achieving food security is contingent upon the ability of all individuals—particularly those with limited income—to access

adequate food at reasonable prices. Therefore, this section analyzes the evolution of fresh fish prices and sardine prices, the latter being one of the most widely produced and consumed fish species in Algerian society, accounting for 60% of total fish production in Algeria in 2021 (Office National des Statistiques, 2024, p. 18)

Fig.1. Evolution of Sardine Prices and the Fresh Fish Price Index during the Period 2010–2022



Source: Compiled by the researcher based on:

1. Office National des Statistiques (ONS), Collections Statistiques, Indice des prix à la consommation, 2010–2023, <https://www.ons.dz/spip.php?rubrique316>
2. Office National des Statistiques (ONS), Indice des prix à la consommation, 2010–2022, <https://www.ons.dz/spip.php?rubrique124>

The preceding figure shows that the Fresh Fish Price Index experienced a continuous increase during the period 2010–2022, rising from 100 to 302. Although the index stabilized at around 210 during the period 2016–2018, it recorded a sharp surge between 2019 and 2020, increasing from 228 to 302 in just three years. Specifically, fresh fish prices rose by 109% in 2018 compared to the base year 2010, but then jumped to 202% by 2022. This increase is attributed to the fluctuation in fish production, which remained stable at an average of 104 thousand tons annually, against the backdrop of sustained demographic growth, as the population increased from 35.978 million in 2010 to 45.631 million in 2022 (Office National des Statistiques, 2024, p. 25), representing a growth of 26.8%.

As for sardines, which are among the most popular dishes in Algerian society and accounted for 60% of total fish production in Algeria in 2021 (Office National des Statistiques, 2024, p. 18), their availability to consumers is closely tied to their prices. The figure shows that sardine

prices rose from 280 DZD to 821 DZD per kilogram, an increase of 193.21% in 2022 compared to 2010.

This sharp price increase is likewise attributed to the fluctuation in fish production, which remained at an average of 104 thousand tons annually, in the context of growing demographic pressure, with the population rising from 35.978 million in 2010 to 45.631 million in 2022 (Office National des Statistiques, 2024, p. 25), marking a 26.8% increase.

4.3.1 Sector Efficiency in Providing Affordable Prices for Individuals:

To assess the efficiency of the sector in providing affordable prices for individuals, it is essential to compare fish prices with the minimum wage, as the latter reflects the income level of low-income groups. Such a comparison helps evaluate the extent to which these groups can secure their basic food needs.

The national minimum wage increased from 15,000 DZD in 2010 to 20,000 DZD in 2022, representing a 30% rise. In contrast, fresh fish prices, according to the Fresh Fish Price Index, increased by 200% over the same period, while sardine prices rose by 193.21%, nearly doubling.

This disparity reflects a significant decline in individuals' purchasing power regarding fish, as wage growth (30%) did not keep pace with the sharp price increase (around 200%). Consequently, the ability of low-income groups to access fish declined markedly during the period (2010–2022).

For instance, if a person is responsible for supporting a family of four, they would need to secure 80 kg of fish annually, based on the global consumption average of 20 kg per capita per year. At a sardine price of 280 DZD per kilogram in 2010, this individual would have had to allocate 22,400 DZD annually ($80 \text{ kg} \times 280 \text{ DZD}$). This amount represented 149% of the national minimum wage at that time (15,000 DZD), i.e., less than one and a half times the wage.

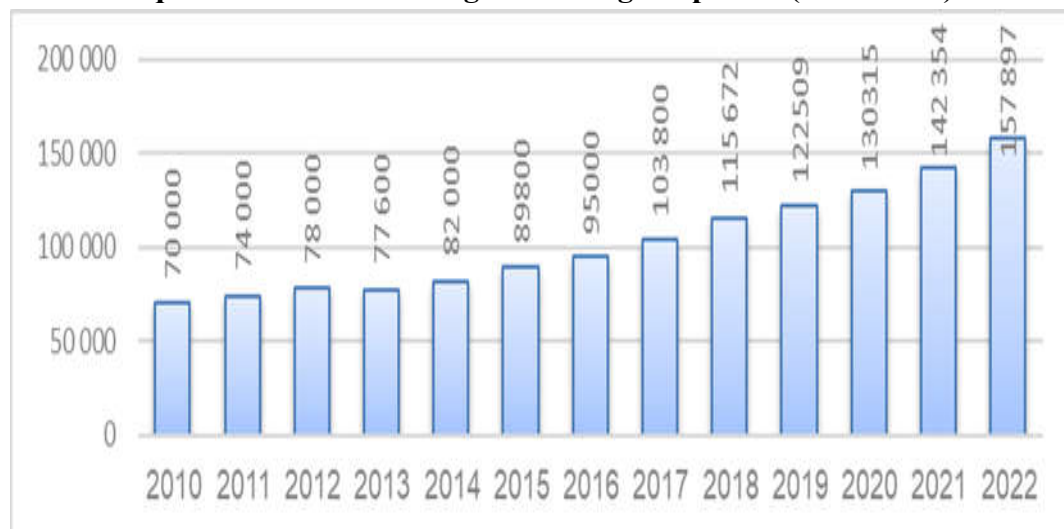
By 2022, however, with sardine prices rising to 821 DZD per kilogram, the same individual would need to allocate 65,680 DZD annually ($80 \text{ kg} \times 821 \text{ DZD}$). This amount is equivalent to 328% of the national minimum wage (20,000 DZD), or more than three times the wage. This

illustrates a sharp decline in individuals' purchasing power, particularly among low-income groups, making it increasingly difficult to obtain sardines in adequate quantities.

Thus, the efficiency of the sector in providing affordable prices that enable individuals to obtain sufficient quantities of fish has been in continuous decline. This is due to the rapid rise in prices, which far outpaced income growth, negatively affecting fish consumption—especially among the most vulnerable groups.

4.4 The Role of the Fisheries Sector in Providing Cash Income for Individuals: Nearly half a billion people worldwide rely on small-scale fisheries as their primary source of livelihood (Food and Agriculture Organization , 2024, p. 151). Accordingly, the fisheries and aquaculture sector is among the most important sectors that generate numerous employment opportunities, given its heavy reliance on manual labor in the production process. The following figure illustrates the number of direct and indirect jobs created in this sector in Algeria.

Fig.2. Evolution of Direct and Indirect Employment in the Fisheries and Aquaculture Sector in Algeria during the period (2010–2022)



Source: Compiled by the researcher based on:

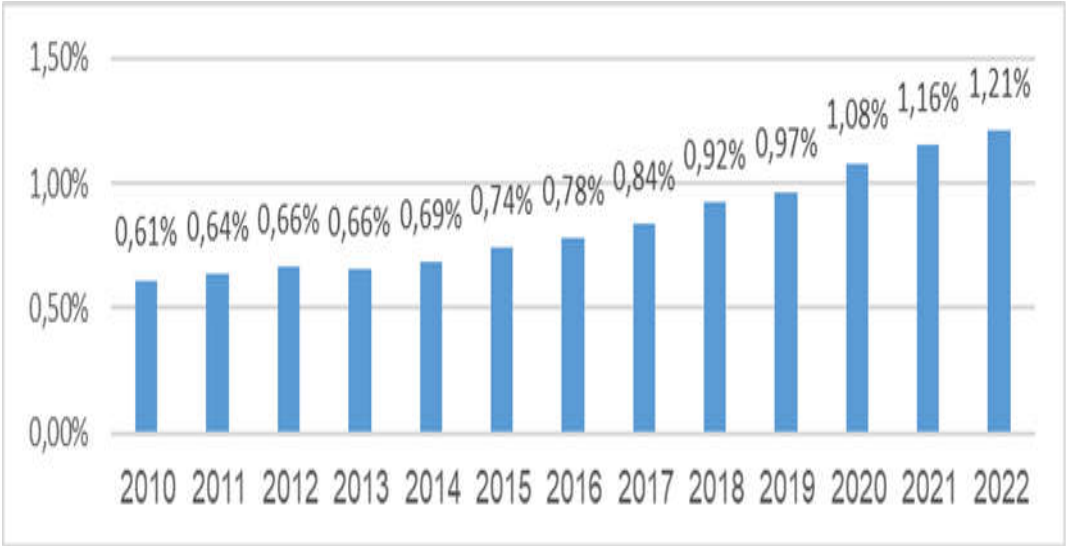
1. ONS, Les principaux indicateurs du secteur de la pêche 2010–2022, website: <https://www.ons.dz/spip.php?rubrique263>

The figure above shows that the fisheries sector has experienced continuous growth in employment generation, as the number of jobs

increased from 70,000 to more than 157,897 during the period 2010–2022, representing a rise of 125.56%. This increase indicates that the sector has nearly doubled the number of available jobs, thereby contributing to injecting more cash income into society. In turn, this cash income enables workers to better meet their living needs, including securing adequate food for themselves and their dependent families. The 2022 annual report of the National Office of Statistics on the key indicators of the fisheries sector reveals that lower-grade professional workers constitute the majority of the workforce in the sector, representing 72% of total employees in 2021, while accounting for 68% of direct employment in 2022 (Office National des Statistiques, 2024, p. 15).

1.4.4 The Efficiency of the Fisheries Sector in Employment and Cash Income Generation within the Economy: To analyze the efficiency of the fisheries sector in job creation, it is necessary to determine its share in the employment of the active labor force in the Algerian economy. The following figure illustrates this:

Fig.3. Evolution of the Contribution of the Fisheries and Aquaculture Sector to Labor Force Employment during the period 2010–2019



Source: Compiled by the researcher based on:

1. World Bank data (<https://data.worldbank.org>)
2. Fig.2

The analysis of Fig.03 shows that the fisheries sector is the least contributing sector in terms of employment opportunities compared to other countries. Despite the slight improvement observed between 2010 and 2022, with its share in employment increasing from 0.61% to 1.21%, it has not reached the levels observed in other regions, where the sector's contribution to employment accounted for 9.9% in African countries and 3.7% in Latin American and Caribbean countries. Therefore, this increase in the sector's contribution to employment in Algeria remains very modest and insufficient to meet the expected aspirations of absorbing more unemployed labor. Moreover, the economic role of the sector remains limited in significantly enhancing individuals' cash incomes, which reduces its capacity to effectively contribute to achieving food security.

5. CONCLUSION

Through our study of the role of the fisheries sector as a model of the blue economy and its contribution to strengthening food security, we conclude that the blue economy plays a significant role in the economies of countries worldwide in achieving food security. It provides more than 20 kg of fish protein per capita annually at the global level and contributes to the generation of income for nearly 500 million people, enabling them to purchase food. This is mainly due to the sector's heavy reliance on labor, which enhances opportunities for accessing additional food sources.

As for the fisheries and aquaculture sector in Algeria, our findings indicate that the sector has not demonstrated efficiency in strengthening food security, whether in terms of providing food in adequate quantities, at affordable prices, or in ensuring sufficient income for a larger number of individuals.

In terms of food provision, the sector achieved an average annual production of 104,000 tons during the study period, equivalent to a per capita share of 2.52 kg of fish. However, the sector failed to achieve allocative production efficiency in enhancing food security for two main reasons:

Insufficient production to provide adequate food: The annual per capita share in Algeria remains significantly below the global average of 20.1 kg, resulting in a food gap of 17.54 kg annually. This means that the

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per capita share of local production is nearly eight times lower than the global average.

Mismanagement of the sector's economic resources: More than 40% of fishing fleet vessels remain underutilized in the production chain, leading to wastage of economic resources and their deterioration due to depreciation and limited use. This has weakened production and restricted investment in available capacities.

With regard to prices, the sector has failed to provide affordable fish prices for low-income groups, which has undermined its contribution to strengthening food security. The price of fresh fish rose by about 200% between 2010 and 2022. Sardines, which represent 60% of fish production in Algeria, witnessed a similar increase. However, this price surge was not matched by a sufficient rise in the minimum wage, which increased by only 30% during the same period. According to the study, in 2022, an individual supporting a family of four had to allocate more than three times the minimum wage to reach the global average per capita fish consumption, compared to less than one and a half times the minimum wage in 2010. This reflects a sharp decline in purchasing power, significantly reducing the ability of low-income households to access fish as a basic food source.

In terms of income generation, the sector managed to double the number of jobs during the study period, rising from 70,000 to over 157,000 direct and indirect positions. Nevertheless, the sector has not been effective in creating sufficient employment opportunities to ensure income for a larger number of individuals, despite its labor-intensive nature. This is mainly due to the sector's weak contribution to overall employment in the economy, with its highest share not exceeding 1.21% of the active labor force. Therefore, it cannot be considered that the sector demonstrates the required efficiency in providing adequate employment opportunities and income to enhance food security.

Based on the findings of this study, several recommendations are proposed to strengthen the role of the fisheries sector and the blue economy in achieving food security in Algeria. This requires increasing the annual domestic production of fish to reduce the existing food gap between Algeria

and the global average, through the development of production techniques, the improvement of fishing methods, the integration of modern technologies, and the modernization of the fishing fleet. Such measures would also help regulate fish prices by increasing supply, thereby ensuring greater affordability for low-income groups. Furthermore, it is recommended to encourage investment in port infrastructure by upgrading and expanding facilities, enhancing storage and distribution capacities, and promoting supportive policies aimed at expanding aquaculture activities. These measures would ultimately enhance the productive and allocative efficiency of the sector within the national economy.

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